

Mackenzie Taylor

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CLINICAL RESEARCHER – QUANTITATIVE FOCUS | BEHAVIORAL DATA SCIENTIST

Versatile and results-driven PhD-trained neuroscience professional, complemented by extensive experience in data science, neuroimaging analysis, and scientific communication. Dedicated to advancing understanding in the field of cognitive neuroscience through innovative research and effective collaboration. Seeking to leverage expertise in research, teaching, and clinical research management to contribute to real-world applications

EDUCATION

PhD in Cognition and Neuroscience

University of Texas at Dallas – August 2023

Dissertation: *Hemodynamic Response Variability and its Relationship to the BOLD Signal in Younger and Older Adults*

MS in Applied Cognition and Neuroscience

University of Texas at Dallas – December 2019

BS in Psychology, Minor in Biology

Texas Wesleyan University – May 2017

TECHNICAL SKILLS

- **Programming & Data Analysis:** R, Python (including machine learning packages, e.g., PySpark), SQL, SPSS, MATLAB, Excel, RMarkdown, Git, cloud-based (Snowflake, Databricks, Azure) and high-performance computing
- **Statistical Methods & Modeling:** Linear and generalized linear regression models, MANOVA, multivariate analysis, clustering, multiple correspondence analysis (MCA), experimental design, hypothesis testing, survey statistics, mixed-methods analysis
- **Data Collection & Management:** Survey design (Qualtrics, SurveyMonkey), data cleaning and validation, behavioral and experimental data pipelines, structured and unstructured data integration
- **Visualization & Reporting:** R- and python-based packages, Excel dashboards, Tableau, reproducible reporting, stakeholder-ready summaries
- **Research & Collaboration:** Study design, cross-functional collaboration, peer-reviewed publication, statistical communication, data storytelling

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

Center for BrainHealth, UT Dallas — Aug 2017 – Nov 2023

- Designed and executed behavioral and neuroimaging studies using rigorous experimental and statistical methodologies to evaluate cognitive task performance
- Led end-to-end research workflows including protocol development, participant recruitment, data collection, statistical modeling, and interpretation
- Analyzed large-scale experimental datasets (300+ participants), modeling task accuracy, efficiency, and perceived effort using multivariate and regression-based techniques
- Applied clustering and demographic stratification to identify performance-based subgroups and behavioral phenotypes
- Built reproducible statistical reports and dashboards in R, SPSS, and MATLAB to support internal decision-making and external publication
- Collaborated with interdisciplinary teams (researchers, clinicians, designers, and industry partners including Jazz Pharmaceuticals) to translate statistical findings into actionable insights
- Integrated quantitative and qualitative data to validate findings and inform iterative study design improvements
- Published three first-author and one second-author journal article and presented peer-reviewed research via seven abstracts at well-respected neuroscience conferences, contributing to evidence-based experimental refinement and digital assessment design

Graduate Teaching Assistant

UT Dallas – School of Behavioral and Brain Sciences — Aug 2017 – Sep 2023

- Supported instruction in statistics, research methods, and experimental design at the undergraduate and graduate levels
- Mentored students in statistical reasoning, data analysis workflows, and reproducible research practices using R and SPSS
- Provided one-on-one guidance on study design, hypothesis testing, and interpretation of results

Certified Pharmacy Technician

Walgreens Pharmacy, Arlington, TX — Jun 2015 – Aug 2017

- Managed sensitive healthcare and prescription data in a regulated, high-volume clinical environment
- Maintained accuracy and compliance while collaborating with pharmacists and healthcare providers
- Developed foundational knowledge of pharmacy claims processing, medication coding, and insurance workflows in a regulated healthcare environment



RESEARCH & ANALYTICAL PROJECT HIGHLIGHTS

- **Participant Stratification via Cluster Analysis:** Applied clustering methods in R and SPSS to identify distinct cognitive performance profiles, supporting subgroup-specific analysis and interpretation
- **Behavioral Task Modeling:** Analyzed task-level engagement and performance metrics to detect efficiency bottlenecks and statistically significant drop-off points
- **Survey Design & Statistical Validation:** Designed and deployed large-scale surveys; implemented data cleaning, validation checks, and statistical quality controls
- **Dashboard & Insight Communication:** Developed R- and Excel-based dashboards to communicate behavioral trends and model outputs to technical and non-technical stakeholders
- **Comparative & Literature-Driven Analysis:** Conducted benchmarking analyses to contextualize study designs and outcomes against published research and industry standards



COMMUNICATION & PUBLICATION HIGHLIGHTS

- **Research Impact:** h-index of 4, i10-index of 3, 58 total citations
- **Scientific Communication:** Presented 7 posters and 2 research talks at major neuroscience conferences
- **Publications:** 3 first-author peer-reviewed journal articles; 1 second-author publication
- **Stakeholder Engagement:** Worked across academic, industry, and healthcare teams to drive research outcomes



ADDITIONAL EXPERIENCE

Owner | Pet Services Business – DFW Area — Dec 2022 – Present

- Scaled a customer-centric service business to 40+ recurring clients with 120+ five-star reviews—applied user experience principles to improve onboarding and retention



CERTIFICATIONS & CLEARANCES

- **Eligible for Top Secret Government Security Clearance**
- **IRB research training and CPR/First Aid certified**
- **Strong understanding of research ethics, data privacy, and inclusive study design**
- **Experience working in remote, collaborative environments to achieve project objectives on strict deadlines**